ESC201

→ Info: → 20 min Oral Exam 60%

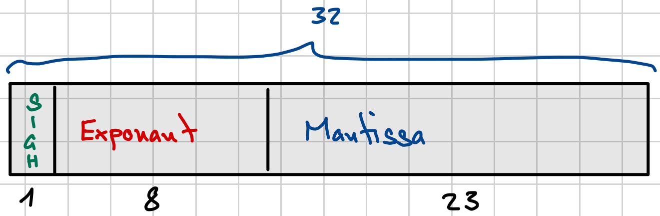
↳ Probably on Monday of last semesterweek

→ every assignment is graded (1 - 6) 40%

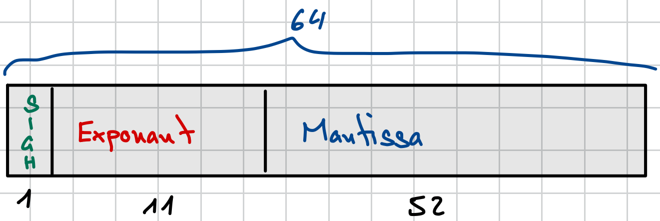
↳ attach .py file to task in teams

IEEE - 754 Standard

→ 32 - Bit float



→ 64 - Bit double



→ Binary : $1.001101 \ldots \times 2^{101}$

$\boxed{\text{Sign}} \times 1.\boxed{\text{Mantissa}} \times 2^{\boxed{\text{Exponant}} - 127}$

Always a One, so not represented

There are weird Bitpatterns inside the Exponant:

$\pm \infty$ / ± 0 / NAN

→ Rounding

$1.\boxed{6}$

9 … 6 } up

5 → Round to nearest even

4 … 1 } down

Mitt. Formel : $ax^2 + bx + c = 0$

I $\quad x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

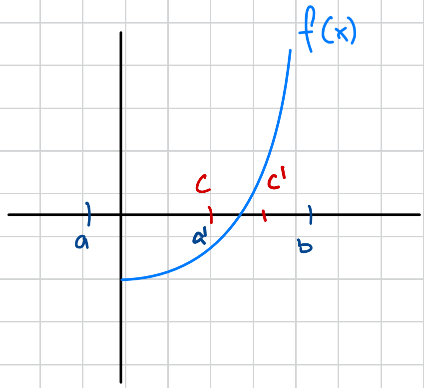
II $\quad x = \dfrac{2c}{-b \pm \sqrt{b^2 - 4ac}}$

↳ $q = -\frac{1}{2}\left(b + \text{sign}(b)\sqrt{b^2 - 4ac} \right)$

↳ $x_1 = \dfrac{q}{a} \qquad x_2 = \dfrac{c}{q}$

root finding : $\boxed{x^x - 100 = 0 \quad \rightarrow \text{ find } x}$ → write in python

$$f(x) = x^x - 100$$



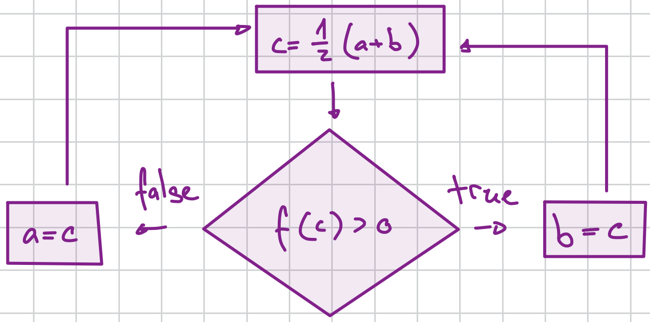
→ $f(a) < 0 \quad \&\& \quad f(b) > 0$

↳ $x \in (a\,;\,b)$

→ split interval in half

↳ if $(\,f(c) > 0\,)\ \{\, b = c \,\}$

else $\{\ a = c\ \}$

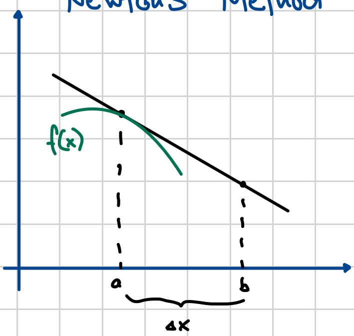


→ until $|a-b| < \varepsilon$

→ or until $\dfrac{|a-b|}{|c|} < \varepsilon$ → relative

→ Slow !

Newton's Method :



$$f(b) \approx f(a) + \Delta x\, f'(a)$$

$$f(b) \approx f(a) + \Delta x\, f'(a) + \frac{\Delta x^2}{2} f''(a)$$

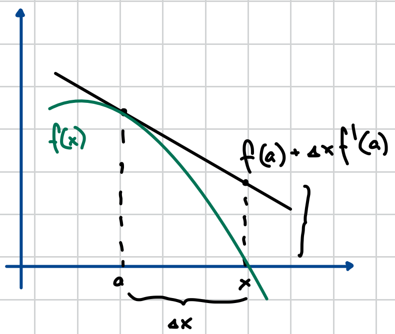
$$\dots$$

} Taylorpolynome

$$\text{Error} = \left[\, \frac{\Delta x^n}{(n-1)!} \int_0^1 (1+t)^{n-1} f^{(n)}(a + t\,\Delta x)\, dt \,\right]$$

→ Optimize Root finding :

We suppose that: $f(x) = 0$



$$\Delta x \cong -\frac{f(a)}{f'(a)}$$

$$x = a + \Delta x \cong a - \frac{f(a)}{f'(a)}$$

Lecture 3

→ Population Growth : → Constant Growth Rate $r = \frac{P_{n+1} - P_n}{P_n}$

↳ $P_{n+1} = (1+r) P_n \rightarrow P_{n+1} = (1+r)(1+r) P_{n-1}$

↳ $P_n(P_0) = (1+r)^n P_0$

→ In reality Populationgrowth is limited by finite ressources

↳ Normalize population by some large sustainable size for population N

p	r
1	0
"small"	→ large & positiv
~ 1	→ small
>1	negative

↳ $p = \frac{P}{N}$ $\qquad r \propto (1-p)$

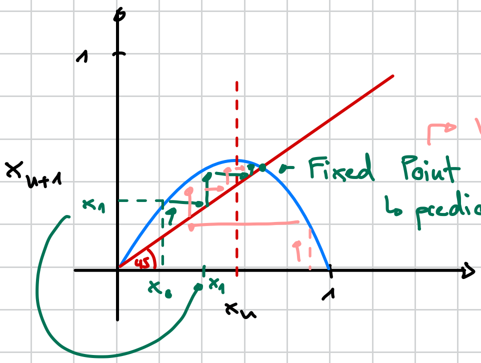
→ Verhulst Model $\quad r = k(1-p)$

↳ $\frac{p_{n+1} - p_n}{p_n} = k(1-p_n)$

↳ $p_{n+1} = p_n + k\, p_n (1 - p_n) \rightarrow \propto p_n^2$!

↳ $x_{n+1} = a\, x_n (1 - x_n) \rightarrow x \in [0; 1]$

Logistic Equation $\qquad a \in [0; 4]$



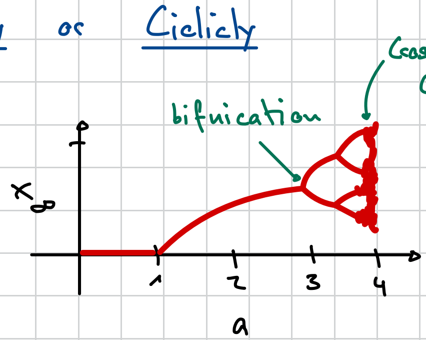
→ We will alway end up here
↳ regardless of starting position

→ Population could also evolve Chaotically or Ciclicly

↳ Depends on Curve

→ Feigenbaum Diagram

↳ Contains "Islands of stability"



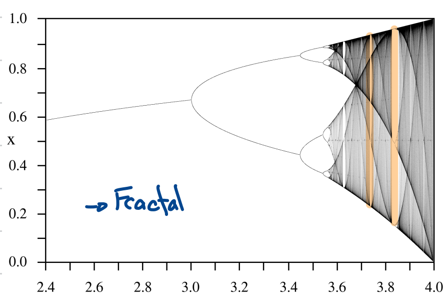
x_∞ → Longtime behaviour
→ x at time ∞

↳ Chaotic Behaviour → Very very similar starting Populations can behave very differently

↳ Feigenbaumpoint → Critical crossover value of a

$a = 3.5 \ldots$

↳ Principles of Bifurication as universal



→ Islands of stability

Lecture 4

→ Complex iterators

$i^2 = -1$, General Complex Number $z = (\underbrace{x}_{\text{Real}} + i\underbrace{y}_{\text{Imaginary}})$

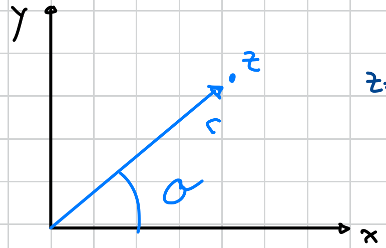
$w = u + iv$

$z = x + iy$

$w + z = (u + x) + i(v + y)$

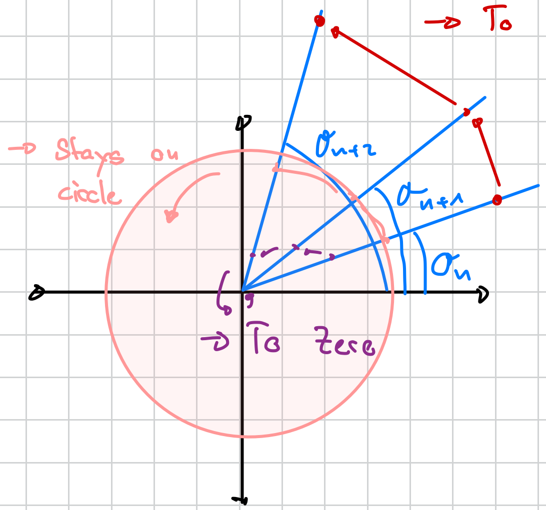
$w \cdot z = x\cdot u + ixv + iyu + i^2 yv$
$= (x\cdot u - y\cdot v) + i(xv + yu)$

$z^2 = (re^{i\theta})^2 = r^2 e^{i(2\theta)}$



$z = re^{i\theta}$

→ Simplest non-linear iteration : $z_{n+1} = z_n^2$



→ Called a Prison
↳ Inside stay inside } Called a Julia-Set

→ In the case of $z_{n+1} = z_n^2$ → Julia set is just "Einheitskreis"

→ With Julia sets, many Fractals can be created

i.e. : $z_{n+1} = z_n^2 + c$

$c = -0.5 + 0.5i$

→ This Julia-Set is connected, not all are

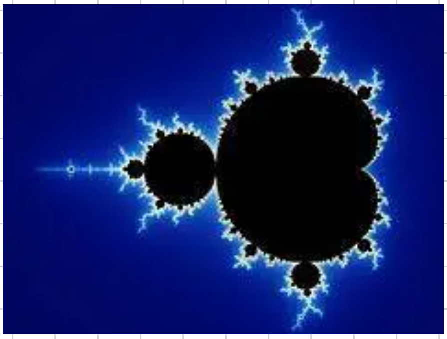
→ If we look at all c and check if it would be connected or not and plot this we would get the Mandelbrot-Set

→ Mandelbrot Set = $\{ c \in \mathbb{C} \mid J_c \text{ is connected} \}$

→ Mandelbrot Set = $\{ z_0 = c \in \mathbb{C} \mid z_{n+1} = z_n^2 + c < \infty \}$

↳ Check if value escapes to ∞ : if $|z_n| > r(c)$ $\{ \notin M \}$

$r(c) = \max(|c|, 2)$



→ Colors are accordint to how many Steps are needed to escape Prison

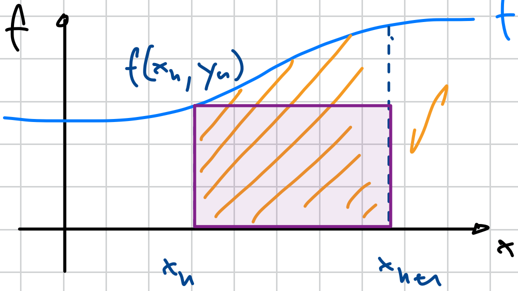
→ For all pixels on screen ~100 iterations

Ordinary Differential Equations:

→ $$\frac{\partial \vec{y}(x)}{\partial \boxed{x}} = \vec{f}(x,y) \qquad \text{unknown: } \vec{y}(x)$$

↳ ordinary since dependent only on x

$$\int_{y_n}^{y_{n+1}} dy = \int_{x_n}^{x_{n+1}} f(x,y)\, dx$$



Approx: $y_{n+1} - y_n \cong f(x_n, y_n) \cdot \overbrace{(x_{n+1} - x_n)}^{\text{Step Size}}$

Truncation Error
↳ comes from Method used,
not from precision in floating point

{ Error depends on Step size

$$\boxed{y_{n+1} = y_n + h \cdot f(x_n, y_n)}$$

→ Forward Euler Method

$$y_n = 0.000123 \; \} \; \dots 32 \; \} \; \dots \; 28$$

Truncation Error — Roundoff Error

ROE shall not grow as big as TE!

→ Error of forward Euler:

$$\begin{aligned} f(x, y(x)) &= f(x, y(x_n + h)) \\ &= f\left(x, y_n + h \frac{\partial y}{\partial x}\Big|_{x_n}\right) \\ &\cong f(x, y_n) + h \frac{\partial y}{\partial x}\Big|_{x_n} \cdot f'(x, y_n) \end{aligned}$$

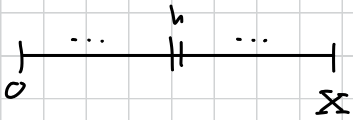
1. Taylor Expansion

2. Taylor Expansion

$$\int_{x_n}^{x_{n+1}} f(x, y(x))\, dx = \int_{x_n}^{x_{n+1}} \left[f(x, y_n) + h \cdot f(x_n, y_n) \cdot f'(x, y_n) \right] dx$$

$$y_{n+1} - y_n \cong h \cdot f(x_n, y_n) + \underbrace{h^2 f(x_n + y_n) f'(x_n, y_n)}_{O(h^2)}$$

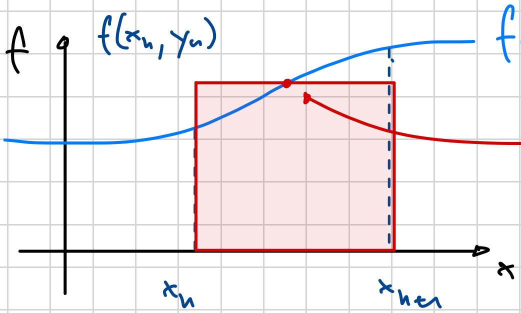
→ Error of a single Step = Local Error



$$N_{steps} = \frac{X}{h}, \quad \text{Global Error} = N_{steps} \cdot O(h^2)$$
$$= O(h)$$

→ Big Error

→ Better Approach



↳ $A = h \cdot f(x_{n+1/2}, \underbrace{y_{n+1/2}}_{\text{Predict}})$

Midpoint method
$$\begin{cases} \text{Prediction}: & y^*_{n+1/2} = y_n + \frac{h}{2} f(x_n, y_n) \\ \text{Corrector}: & y_{n+1} = y_n + h \cdot f(x_{n+1/2}, y^*_{n+1/2}) \end{cases}$$

↳ 2 evaluations of Right hand side of ODE
↳ more expensiv calculations

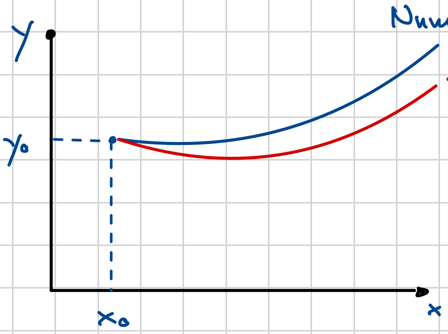
→ Local Error $= O(h^3)$

→ Global Error $= O(h^2)$

Another ODE Example : $\frac{\partial^2 y}{\partial x^2} + q(x) \frac{\partial y}{\partial x} = r(x)$

↳
$$\boxed{\begin{aligned} \frac{\partial y}{\partial x} &= z(x) \\ \frac{\partial z}{\partial x} &= r(x) - q(x) \cdot z(x) \end{aligned}}$$
ODE

→ In general :
$$\frac{\partial y_i}{\partial x} = f_i(x, y_0, y_1, \dots y_{n-1})$$



$$y_0 = y(x_0)$$

Predator - Prey Behaviour

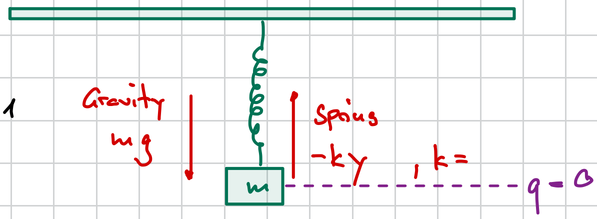
Lecture 5

→ Symplectiv Integrators

↳ Forward Euler
↳ Runga-Kunta

} Black Box Methods → Not always optimal

→ Example : Harmonic Oscilator :



$F = mg - ky$, $m=1$, $k=1$

$\downarrow$

$F = g - y$ / $q = -F = y - g$

→ Newtons law : $F = ma = m\frac{d^2y}{dt^2} = m\ddot{y}$

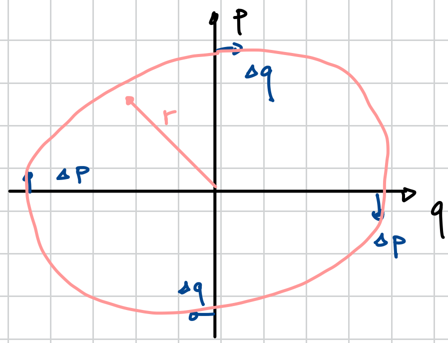
→ $\frac{dy}{dt} = \frac{dq}{dt} = y$

→ $\ddot{q} = \ddot{y} = F$ $\boxed{\ddot{q} = -q}$ → 2nd order ODE

→ Transform to two complete 1st order ODE

$p = mv = \dot{m}\dot{v} = \dot{q}$ $\begin{cases} q = \dot{p} \\ \dot{q} = \ddot{p} \end{cases}$

↳ $\boxed{\begin{matrix} \dot{p} = -q \\ \dot{d} = f \end{matrix}}$ → Hamiltonian equations for Harmonic Oscilator



→ Phase Space Diagram

$r^2 = q^2 + p^2$

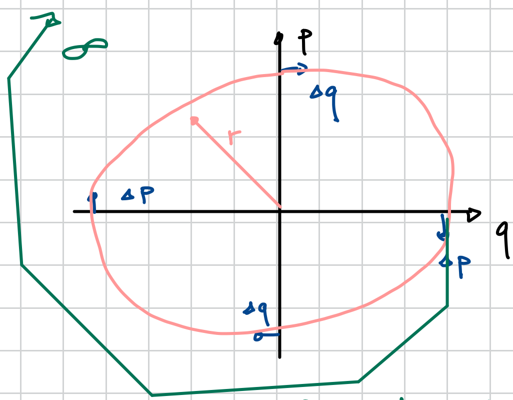
$\boxed{\text{"Energy"} = q^2 + p^2}$ → conserved

Using Forward Euler:

$q_{n+1} = q_n + h\, p_n$
$p_{n+1} = p_n - h\, q_n$
} Solved simultaniously

$$\boxed{\begin{aligned} q_{n+1}^2 + p_{n+1}^2 &= (1+h^2)(q_n^2 + p_n^2) \\ r_{n+1}^2 &= (1+h^2)\, r_n^2 \end{aligned}}$$

↳ Exponentially divergent



forward Euler

Alternativ :

$E = \underset{\substack{\text{Kin.}\\\text{Energy}}}{T} + \underset{\substack{\text{Pot.}\\\text{Energy}}}{U} = \frac{1}{2}mv^2 + \frac{k}{2}y^2$

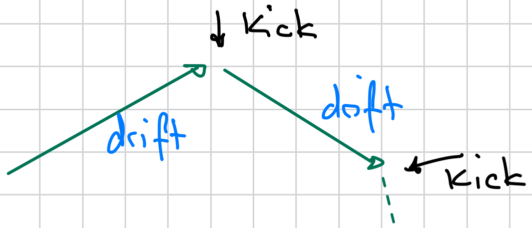
$= \frac{1}{2}(p^2 + q^2)$

$$H = \frac{1}{2}(p^2+q^2) = T(p) + U(q)$$ → Problem is seperable

$$\boxed{\dot{q} = \frac{\partial H}{\partial p} \qquad \dot{p} = \frac{\partial H}{\partial q}}$$

→ $T(p)$ = Free Motion ("Drift Operator")

→ $U(q)$ = Momentum Update ("Kick")



Leapfrog Method:

$h = \Delta t$ → Timestep

$$\boxed{\begin{aligned} q_{n+1/2} &= q_n + \tfrac{1}{2} h p_n \\ p_{n+1} &= p_n - h\, q_{n+1/2} \\ q_{n+1/2} &= q_{n+1/2} + \tfrac{1}{2} h\, p_{n+1} \end{aligned}}$$

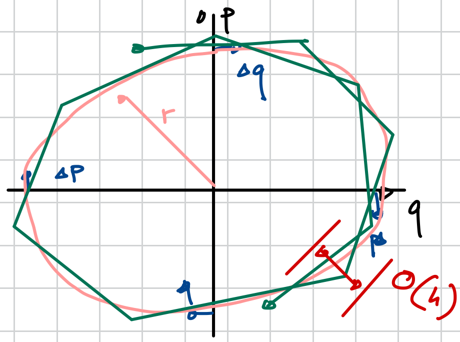
→ Half Drift
→ Full Kick
→ Half Drift

→ Rewrite in terms of position and velocity

x_0 & v_0 → initial conditions

$$\boxed{\begin{aligned} \vec{x}_{1/2} &= \vec{x}_0 + \tfrac{1}{2} h \vec{v}_0 \\ \vec{v}_1 &= \vec{v}_0 + h\left(-\nabla u(\vec{x}_{1/2})\right) \\ \vec{x}_1 &= \vec{x}_0 + 1/2\, h\, \vec{V}_1 \end{aligned}}$$

$$\nabla u = \frac{\partial u}{\partial x}\hat{\imath} + \frac{\partial u}{\partial t}\hat{\jmath} + \frac{\partial u}{\partial t}\hat{k}$$

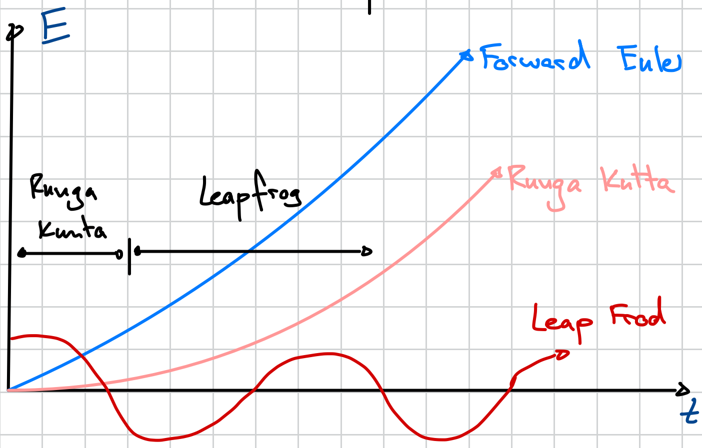


→ Does not derive away

→ There is a local error
↳ Error accumulates in the angle

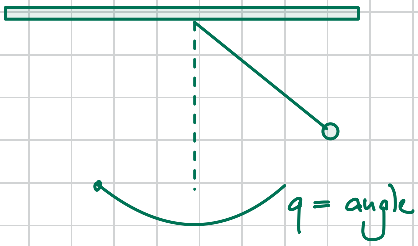
Numerical Period = $\tilde{P}$erion ≠ Period

$$\tilde{H} \neq H$$



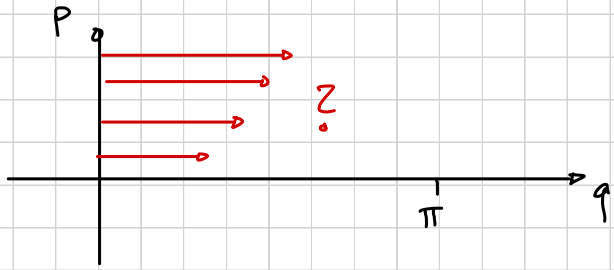
Exercises 4

→ Simple Pendulum

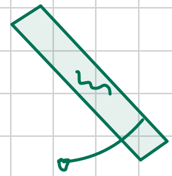


$$H = \frac{1}{2} p^2 - \varepsilon \cos(q)$$

→ make phase space plots



→ Real Pendulum



→ Test and compare:

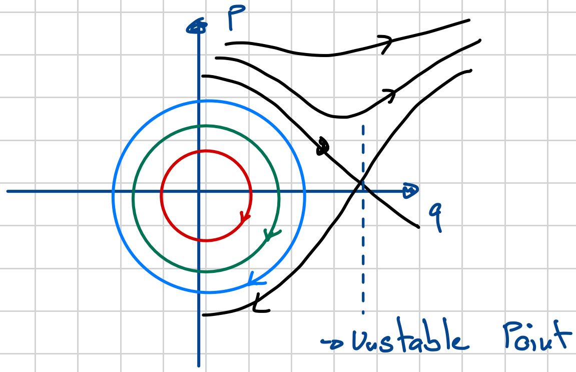
- → leapfrog
- → midpoint
- → Runga Kutta 4

Lecture 6

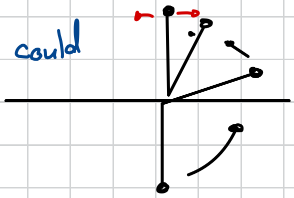
$H = T + U$ → Hamiltonian, separable

T: Kin Energy

U: Potential Energy



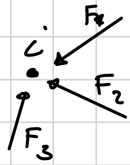
→ Pendulum could fall in either side



Uranus "Clock" of Solarsystem

→ $N = 9$ → 8 planets + 1 Sun

↳ Sun moving arround as well



(N-1) Total forces on body i

→ $N(N-1)$ total forces

↳ Use Newtons 3'rd

→ # of forces : $F_{ij} = -F_{ji}$

$$\rightarrow \#F = \frac{N(N-1)}{2} \rightarrow \mathcal{O}(N^2)$$

→ Example $N = 10^{12}$ → $N_{\text{interactions}} = \frac{10^{24}}{2} \times \sim 20$ Flops (Floating Point Operations)

→ $N_{\text{Flops}} = 10^{25}$

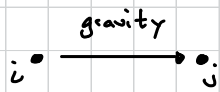
→ $N_{\text{Sec per Year}} = 10^{7.5}\,s$

→ $\frac{10^{25}}{10^{18}} = 10^7$ seconds $\approx \frac{1}{3}$ Year

Big Computers: Available today
- Gigaflops $= 10^9\,s^{-1}$
- Teraflops $= 10^{12}\,s^{-1}$
- Petaflops $= 10^{15}\,s^{-1}$
- Exaflops $= 10^{18}\,s^{-1}$

→ Possible in a few seconds $\quad \mathcal{O}(n) \rightarrow \mathcal{O}(n \log n) \rightarrow \mathcal{O}(n)$

$$\rightarrow F_{ij} = \underbrace{\frac{G m_i m_j}{|\vec{r}_i - \vec{r}_j|^2}}_{\text{Distance}} \cdot \frac{(\vec{r}_j - \vec{r}_i)}{|\vec{r}_j - \vec{r}_i|}$$



$$= \frac{G m_i m_j}{|\vec{r}_i - \vec{r}_j|^2}(\vec{r}_j - \vec{r}_i)$$ → Force on i due to j

$$\vec{F}_i = \sum_{j \neq i} \frac{\boxed{G} m_i m_j}{|\vec{r}_i - \vec{r}_j|^3}(\vec{r}_j - \vec{r}_i)$$

G: one of the least known constants

↳ known to ~ 5 decimal points

↳ Also influences, how exact we know Mass of planets

→ G & M_{Sun} → really unknown

→ $G M_{\text{Sun}} = K^2 =$ Gaussian Constant → known pretty good ~ 15 decimal points

$$\vec{F}_i = \sum_{j \neq i} \frac{k^2 m_i m_j}{|\vec{r}_i - \vec{r}_j|^3} (\vec{r}_j - \vec{r}_i)$$

Unit

$m_i \, [M_{Sol}]$

$r_i \, [AU]$ → Astronomical Unit

$t \, [days]$

$$\dot{\vec{p}} = -\frac{\partial H}{\partial \vec{q}} = \frac{\partial u(\vec{q})}{\partial \vec{q}} = -\nabla u$$

$$\vec{a}_i = \frac{\vec{F}_i}{m_i} = -\nabla u$$

→ Leapfrog

$\forall i$

→ Drift: $\vec{r}_{i,n+\frac{1}{2}} = \vec{r}_{i,n} + \frac{h}{2} \vec{v}_{i,n}$

→ Kick: $\vec{v}_{i,n+1} = \vec{v}_{i,n} + h \, \vec{a}_i \big(\overbrace{\{ \vec{r}_{i,n+1/2}, \vec{r}_{j,n+1/2}, \dots \}}^{\text{All body Positions}} \big)$

→ Drift: $\vec{r}_{i,n+1} = \vec{r}_{i,n+1/2} + \frac{h}{2} \vec{v}_{i,n+1}$

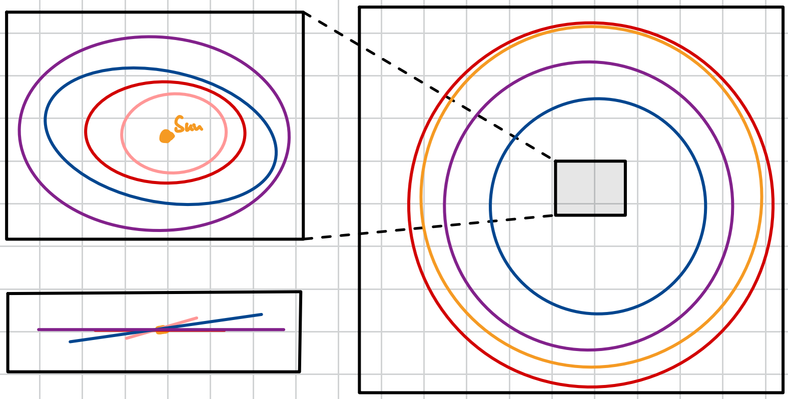
→ Apropriate Timestep: ~ 3 days → $h \,\hat{=}\, 3$

→ Starting Positions will be given with task

→ $\vec{r}$ in $[AU]$
$\vec{v}$ in $[\frac{AU}{day}]$

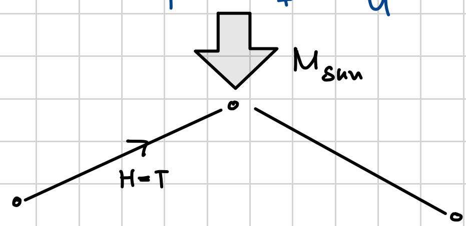
m in $[M_{Sun}]$
t_{ic} in [Julian Date]
Δt in 3 days

→ Plot :

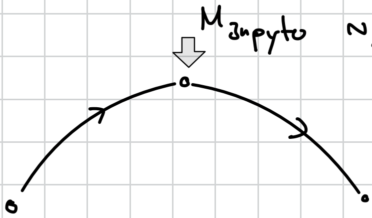


→ $H = H_{Keppler} + \varepsilon H_{Pert}$

$T + U$



→



Partial Differential Equations

→ Multiple Derivatives

→ No "Black Box" method of solving all of them

Diff. Solving Methods

$$\boxed{\frac{\partial^2 u}{\partial^2 t} = v^2 \frac{\partial^2 u}{\partial x^2}}$$

→ One Dimensional Wave Equation
→ Hypobolic PDE's

$$\boxed{\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}}$$

→ One Dimensional Diffusion Equation
→ Parabolic PDE's

$$\boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \overbrace{p(x,y)}^{\text{given}}}$$

→ Another Class would be: Eliptic PDE
↳ No time but two spacial derivatives

→ wanted: $u(x,y)$

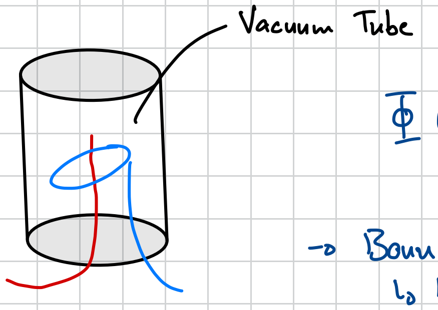
→ $p(x,y) = 0$ → Laplace Equation

$$\boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0}$$

$$\nabla^2 u = 0$$

→ One of things describable:

→ Electrostatic Potential in Vacuum



$\Phi(x,y)$

→ Boundary Conditions
↳ Determine Solution

$$\nabla^2 \Phi = 0$$

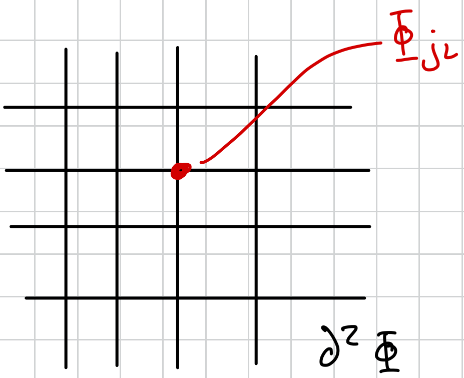
Step ① → Discretize (Make

Grid Spacing Δx, Δy

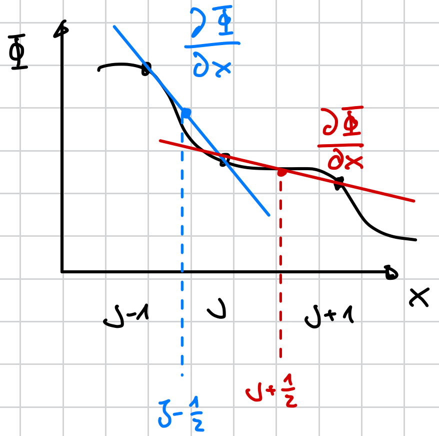
$x_j = x_0 + j\Delta x$, $j = 0, 1, \dots, J$

$y_l = y_0 + l\Delta y$, $l = 0, 1, \dots, L$

→ Simplify: $J = L = N$ $\quad \Delta x = \Delta y = 1$



$$\frac{\partial^2 \Phi}{\partial x^2} = ?$$



$$\frac{\partial^2 \Phi}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial \Phi}{\partial x}\right)$$

$$\left.\frac{\partial \Phi}{\partial x}\right|_{j-\frac{1}{2}} \cong \frac{\Phi_j - \Phi_{j-1}}{\Delta}$$

$$\left.\frac{\partial \Phi}{\partial x}\right|_{j+\frac{1}{2}} \cong \frac{\Phi_{j+1} - \Phi_j}{\Delta}$$

$$\frac{\partial}{\partial x}\left[\frac{\partial \Phi}{\partial x}\right] \cong \frac{\left.\frac{\partial \Phi}{\partial x}\right|_{j+\frac{1}{2}} - \left.\frac{\partial \Phi}{\partial x}\right|_{j-\frac{1}{2}}}{\Delta}$$

$$\cong \frac{\Phi_{j+1} - 2\Phi_j + \Phi_{j-1}}{\Delta^2} \quad \Big\} \text{ Stencil}$$

$$\cong \frac{1}{\Delta^2}\begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$$

$$\left.\frac{\partial^2 \Phi}{\partial y^2}\right|_{l} \cong \frac{1}{\Delta^2}\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$\left[\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2}\right]_{j,l} \cong \frac{1}{\Delta^2}\begin{bmatrix} & 1 & \\ 1 & -4 & 1 \\ & 1 & \end{bmatrix}$$

$\nabla^2 \Phi \overset{?}{=} 0$ → stated by Equation

$\frac{1}{\Delta^2}\begin{bmatrix} & \circ & \\ \circ & \circ & \circ \\ & \circ & \end{bmatrix} \Phi = 0$ → Discrete Statement

$$\forall_{j,l} : \quad \phi_{j+1,l} + \phi_{j-1,l} + \phi_{j,l+1} + \phi_{j,l-1} - 4\phi_{j,l} = 0$$

$$\phi_{jl}^{(new)} = \frac{1}{4}\begin{bmatrix} & \circ & \\ \circ & \bullet & \circ \\ & \circ & \end{bmatrix} \phi^{(old)}$$

→ One has to start with guesses

Iterate many times

↳ Speed of computation depends of quality of guess

→ Converges very slowly → take a long time

error = |old − new|

↳ On a N×N grid, reduce error a factor of 10^p, it takes

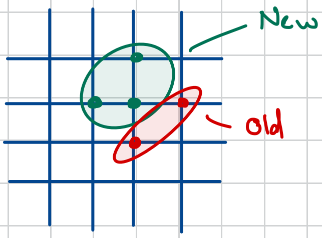
$$N_{iter} \cong \frac{1}{2} p N^2$$

→ N^2 grid points to calculate at each iteration:

$$N_{ops} = \underbrace{N_{ito}}_{\frac{1}{2} p N^2} \cdot N^2 \cdot \underbrace{\frac{N_{ops}}{point}}_{4}$$

$$N_{ops} = 2 p N^4 \rightarrow O(N^4)$$

Gauss - Seidel - Sweeps



→ Faster

Successive Over-Relaxation (SOR)

(There are Methods that $\underbrace{O(\log N N^2)}_{\text{FFT's}}$ & $\underbrace{O(N^2)}_{\text{Multigrid}}$)

→ $O(N^3)$

$$\Phi^{new}_{j,l} = \Phi^{old}_{j,l} + \frac{1}{4} \underbrace{\left[\text{(stencil)} - 4\,\Phi^{old}_{j,l} \right]}$$

$$\Phi^{new}_{j,l} = \Phi^{old}_{j,l} + \frac{\omega}{4} \left\{ \text{(stencil)} \right\} \Phi^{old}$$

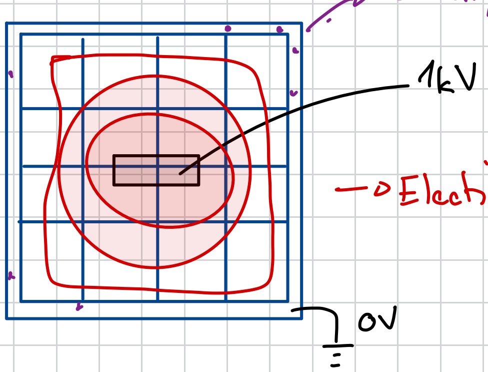
$\omega = 1$ → Jacobi

$\omega < 1$ → worse

$\omega > 1 \,\|\, \omega < 2$ → Successive Over Relaxation → Determine Experimentally

$\omega \geq 2$ → Unstable

N	N_{ito} 0.1% relative error	$\omega_{optimal}$
10		
20		
50		

→ 

→ Boundary Condition → Has to stay the same

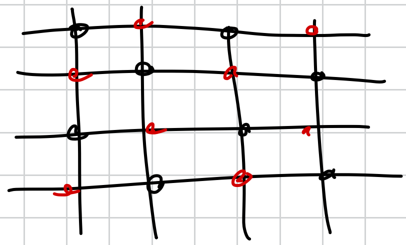
$$\rightarrow \Phi^{n+1}_{j,l} = \Phi^{n}_{jl} + R_{jl} \cdot \left(\text{(stencil)} \right)$$

↳ 0 if BC

↳ $\frac{\omega}{4}$ else

→ Electric Field → Gets more and more "Boxy"

→ In Place: one grid of Φ



→ Black points only depend on red and vise versa

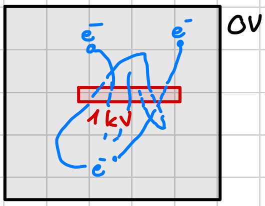
One Iteration : → sweep through all black

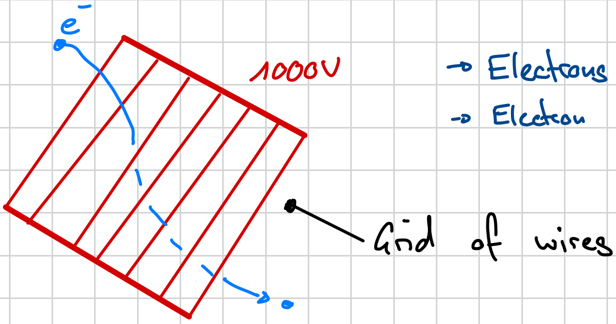
→ sweep through all red

→ Parallel gains

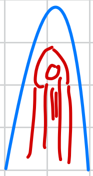
Lecture 8 Interpolation and Electron Beams

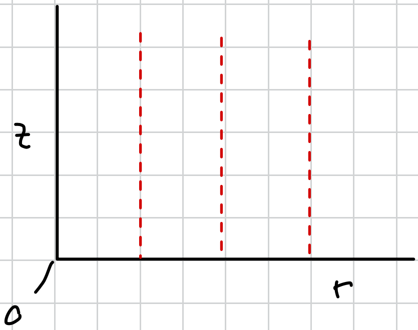
Prev. Exercise:





→ Electrons can pass
→ Electron is attracted by positiv Voltage





$\nabla^2 \phi = 0$ Cylinder Coord.

→ We will look at system in 2d

$$\vec{F} = m_e \vec{a}$$

Accelaration of e^-

$-\nabla^2 \phi$
PDE

Electrons Charge
Electric Field
Magnetic Field

$$\vec{F} = e\left(\vec{E} + \cancel{v_e \times B}\right), \qquad \vec{E} = -\nabla\phi$$

$$a = \left(\frac{e}{m_e}\right)\left(-\nabla \phi\right)$$

$$\frac{\partial^2 x}{\partial t^2} \qquad \ddot{x} \equiv \dot{v}_x = \left(\frac{e}{m_e}\right)\left(\frac{\partial \phi}{\partial x}\right)$$

$$\frac{\partial y}{\partial t} \qquad \dot{v}_y = \left(\frac{e}{m_e}\right)\left(\frac{\partial \phi}{\partial y}\right)$$

$$\dot{x} = v_x$$

$$\dot{y} = v_y$$

→ System of 4 coupled 1st order ODE's

Leapfrog, Runge Kutta IV, ...

Units:

$e = 1.6 \times 10^{-19}$ C

$m_e = 9.11 \times 10^{-31}$ kg

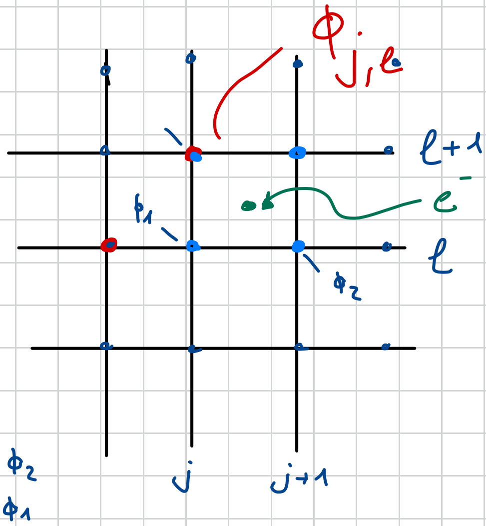
$\frac{e}{m_e} = 1.76 \times 10^{11}$ C/kg

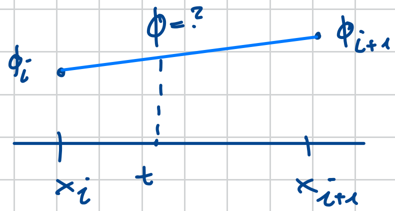
$$\Phi = \left[\frac{Nm}{C}\right] \qquad \nabla\Phi = \left[\frac{N\cancel{m}}{C\cancel{m}}\right]$$

$$\left(\frac{e}{m_e}\right)\left(-\nabla\phi\right) = \left[\frac{C}{kg}\right]\cdot\left[\frac{N}{C}\right] = \left[\frac{N}{kg}\right] = \left[\frac{\cancel{kg}\, m s^{-2}}{\cancel{kg}}\right] = \left[m s^{-2}\right]$$

$\phi(x,y) = ?$ $\phi_{j,l}$

→ Interpolation since e^- not exactly on grid



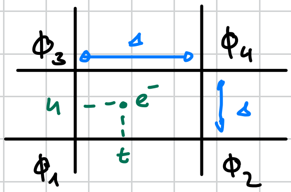


$\phi = t\,\phi_2 + (1-t)\,\phi_1$

→ If $t=1$ → $\phi = \phi_2$

→ If $t=0$ → $\phi = \phi_1$

$u,t \in [0,1)$ → because in numerics $1 \neq 1$



$u = 0 \rightarrow \quad \phi(x, y_l) = t\,\phi_2 + (1-t)\,\phi_1$

$u = 1 \rightarrow \quad \phi(x, y_{l+1}) = t\,\phi_4 + (1-t)\,\phi_3$

$$\phi(x,y) = (1-u)\left[(1-t)\,\phi_1 + t\,\phi_2\right] + u\left[(1-t)\,\phi_3 + t\,\phi_4\right]$$

$$\boxed{\phi(x,y) = (1-u)(1-t)\,\phi_1 + (1-u)\,t\,\phi_2 + u(1-t)\,\phi_3 + u\,t\,\phi_4}$$

↳ Bilinear Interpolation

↳ Continuous → we can take derivative

$$x \rightarrow t = \frac{x - x_j}{x_{j+1} - x_j} = \frac{x - x_j}{\Delta}$$

$$u = \frac{y - y_l}{y_{l+1} - y_l} = \frac{y - y_l}{\Delta}$$

$$\left.\frac{\partial \phi}{\partial x}\right|_u = \frac{\partial t}{\partial x} \cdot \frac{\partial \phi}{\partial t} = \frac{1}{\Delta}\left[-(1-u)\,\phi_1 + (1+u)\,\phi_2 - u\,\phi_3 + u\,\phi_4\right]$$

$$= \frac{1}{\Delta}\left[(1-u)(\phi_2 - \phi_1) + u(\phi_4 - \phi_3)\right]$$

$$\left.\frac{\partial \phi}{\partial y}\right|_t = \frac{\partial t}{\partial x} \cdot \frac{\partial \phi}{\partial u} = \frac{1}{\Delta}\left[-(1-t)\,\phi_1 - t\,\phi_2 + (1-t)\,\phi_3 + t\,\phi_4\right]$$

$$= \frac{1}{\Delta}\left[(1-t)(\phi_3 - \phi_1) + t(\phi_4 - \phi_2)\right]$$

$$\dot{x} = v_x$$

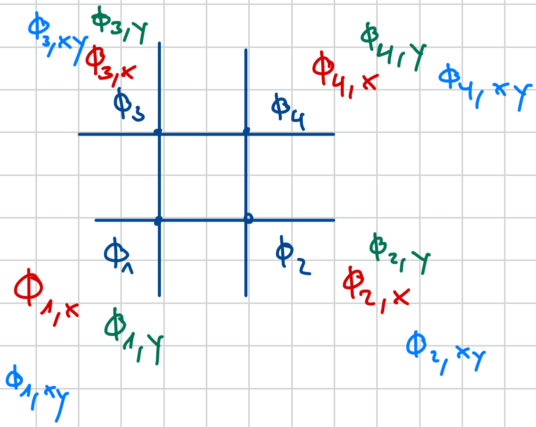
$$\dot{v}_x = \left(\frac{e}{m_e}\right)\left[-\frac{1}{\Delta}\left((1-u)(\phi_2 - \phi_1) + u(\phi_4 - \phi_3)\right)\right]$$

$$\dot{y} = v_y$$

$$\dot{v}_y = \left(\frac{e}{m_e}\right)\left[-\frac{1}{1}\left(\left(1-t\right)\left(\phi_3-\phi_1\right)+t\left(\phi_4-\phi_2\right)\right)\right]$$

→ Use solver f.e. Leapfrog

(→) For Bicubic Intepolation: Use hermantic functions for Mixing



$$\equiv \frac{\partial^2 \phi}{\partial x \partial y}\bigg|_1$$

$$\equiv \frac{\partial \phi}{\partial x}\bigg|_1$$

$$\phi(x,y) = \sum_{i=0}^{3}\sum_{j=0}^{3} a_{ij}\cdot t^i u^j$$

$$\begin{bmatrix} a_{00} & a_{01} & \dots & \\ a_{10} & a_{11} & & \\ \dots & & a_{22} & \\ & & & a_{33} \end{bmatrix} = M^T \cdot \left[\begin{array}{cc|cc} \phi_1 & \phi_2 & \phi_{1y} & \phi_{2y} \\ \phi_3 & \phi_4 & \phi_{3y} & \phi_{4y} \\ \hline \phi_{1x} & \phi_{2x} & \phi_{1,xy} & \phi_{2xy} \\ \phi_{3x} & \phi_{4x} & \phi_{3xy} & \phi_{4xy} \end{array}\right] \cdot M$$

$$M = \begin{bmatrix} 1 & 0 & -3 & 2 \\ 0 & 1 & 3 & -2 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

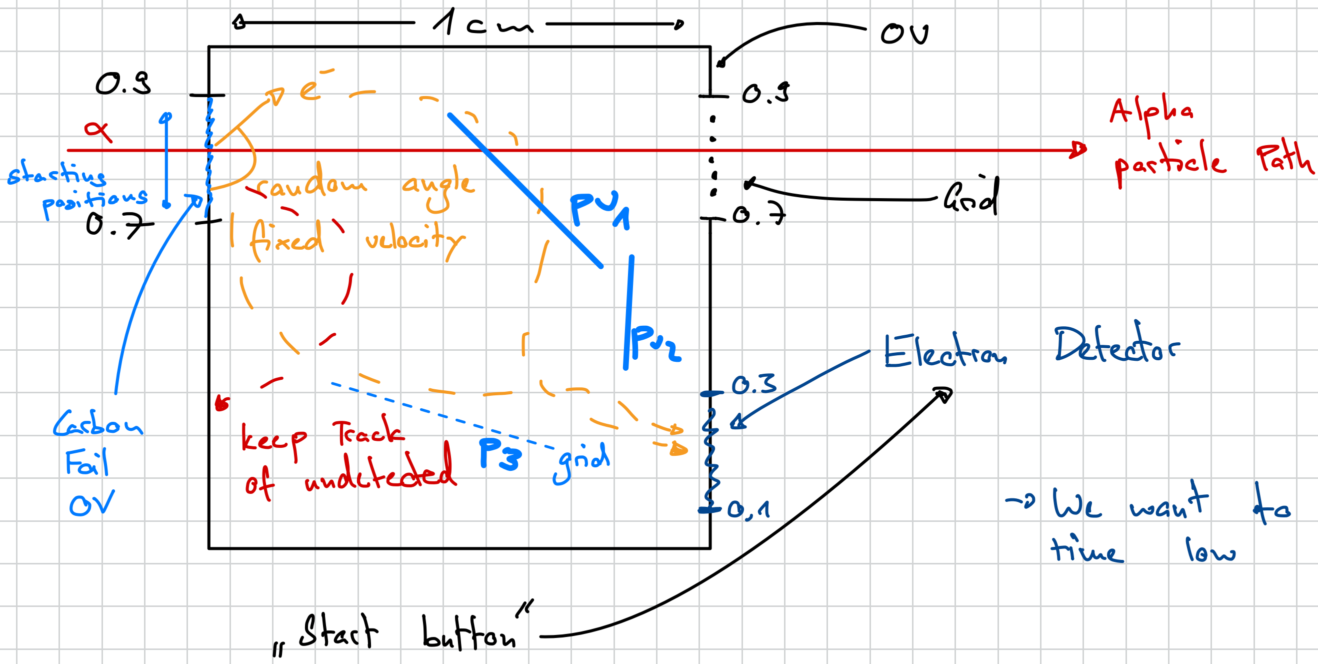
(→ Bicubic Intepolation should be smoothe than Bilinear)

$$\phi(t,u) = \begin{bmatrix} 1 & t & t^2 & t^3 \end{bmatrix} \cdot \begin{bmatrix} a_{i,j} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ u \\ u^2 \\ u^3 \end{bmatrix}$$

2 Weeks to the Prize :

→ week (1) : Code validation

→ week (2) : Design Phase

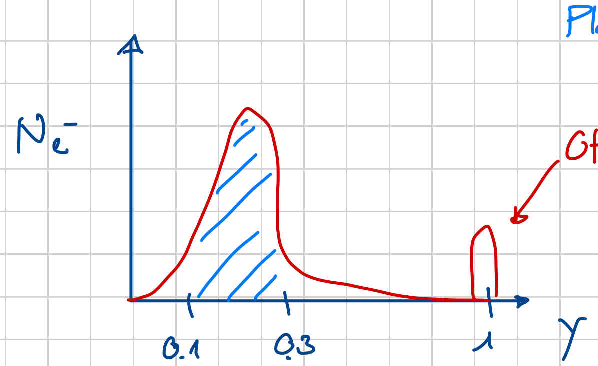


→ We want to keep time low

→ Quality assessment :

$V_e = 10^6 \, m s^{-1}$

Plate Voltage = [-1000, 1000]

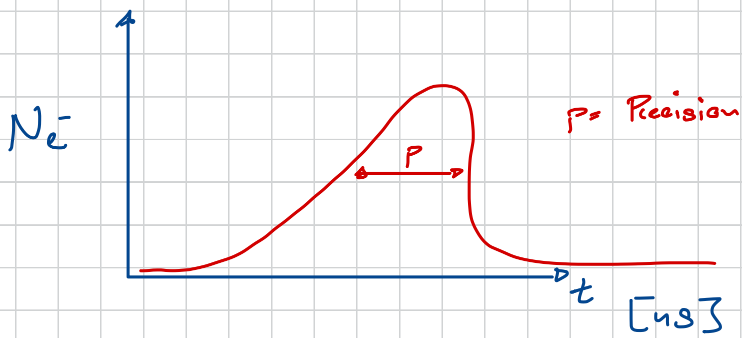


Other wall electrons

$\Phi(x,y) = 1000 \, V$

$R_{je} = 0$

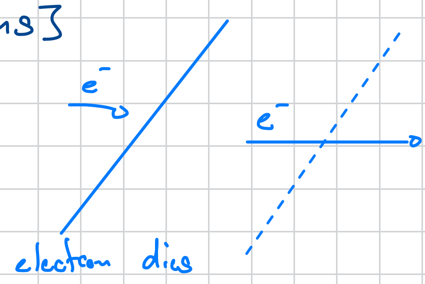
↳ No charge in S.O.R.



p = Precision of device

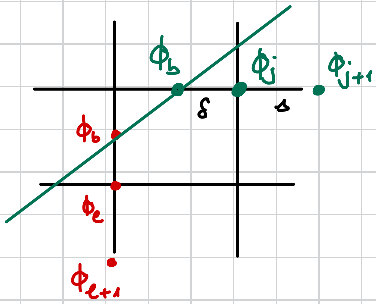
max # of elmts : 10

"Draw Boundary Conditions"

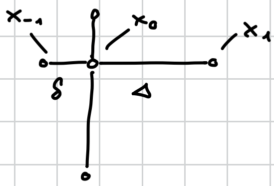


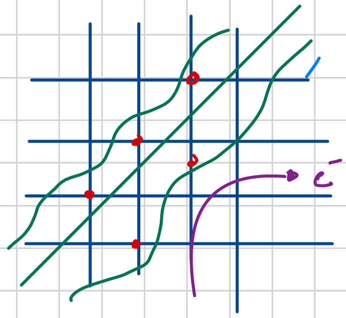
Simulating Electrons Extra

→ Coming up with a stencil   o—o—o $\nabla^2 \phi = 0$



→ Asymetrical Stencil





$$\phi(x) = ax^2 + bx + c$$

$$\phi'(x) = 2ax + b$$

$$\phi''(x) = 2a \qquad \nabla^2 \phi \simeq \frac{\partial^2 \phi}{\partial x^2}$$

Boundry Conditions:

$$\phi(0) = \phi_j \implies c = \phi_j$$

$$\phi(1) = \phi_{j+1} \implies a + b + \phi_j = \phi_{j+1}$$

$$b = \phi_{j+1} - \phi_j - a$$

$$\phi\left(-\frac{\delta}{\Delta}\right) = \phi_b \implies a \cdot \left(\frac{\delta}{\Delta}\right)^2 - b\left(\frac{\delta}{\Delta}\right) + \phi_j = \phi_b$$

$$a\left(\frac{\delta}{\Delta}\right)^2 - \left(\phi_{j+1} - \phi_j - a\right)\left(\frac{\delta}{\Delta}\right) + \phi_j = \phi_b$$ → solve for a

$$a\left(\frac{\delta^2}{\Delta^2} + \frac{\delta}{\Delta}\right) = \phi_b + \frac{\delta}{\Delta}\left(\phi_{j+1} - \phi_j\right) - \phi_j$$

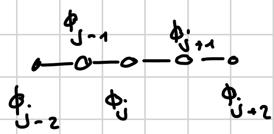
$$a = \frac{\Delta}{\delta} \cdot \frac{\left(\phi_b - \phi_j + \frac{\delta}{\Delta}\left(\phi_{j+1} - \phi_j\right)\right)}{\left(\frac{\delta}{\Delta} + 1\right)}$$

→ $\delta = \Delta$

↳ $\phi_{j-1} - 2a\phi_j + \phi_{j+1}$

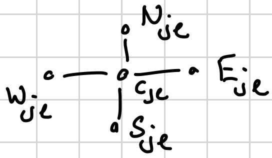
$$2a = \frac{\frac{\Delta}{\delta}\phi_b - \left(\frac{\Delta}{\delta} + 1\right)\phi_j + \phi_{j+1}}{\frac{1}{2}\left(\frac{\delta}{\Delta} + 1\right)}$$

$\phi''(x) = 2a$ → 4th order polynomial:



→ We had extra array with weights $R_{je} < \begin{matrix} \frac{\omega}{4} \\ 0 \end{matrix}$

→ We could also use more weights.



$C_{je} = 4$

$S, E, N, W = 1$

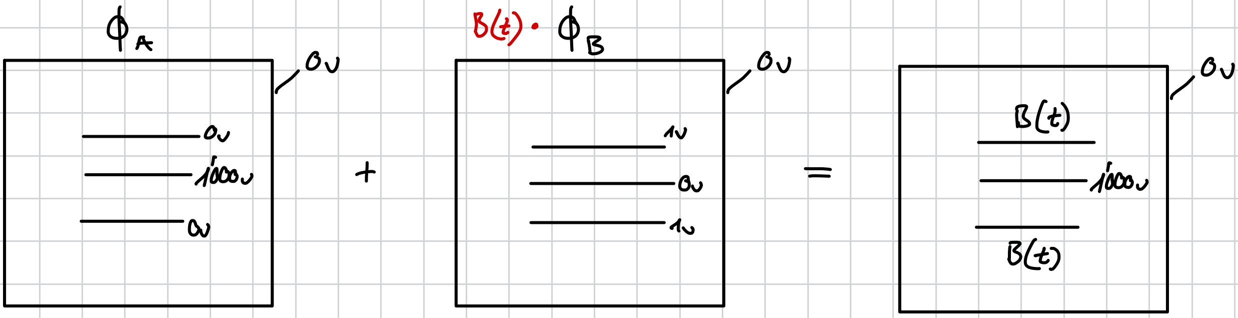
→ Timevariable Voltage Elmt

→ Potential is Linear

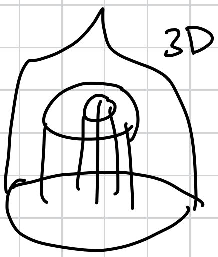
if ϕ_A is a Solution to $\nabla^2 \phi_A = 0$
and ϕ_B is a solution to $\nabla^2 \phi_B = 0$

then $\phi_A + \phi_B$ is also a solution

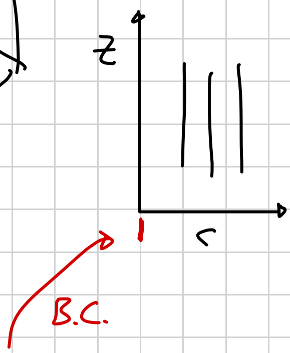
$A\phi_A + B\phi_B$ can be arbitrary contants $A(t)$, $B(t)$



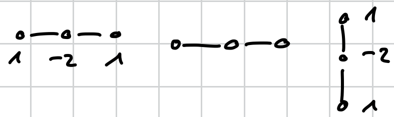
Vacuum Tube



$$\nabla^2 \phi = \underbrace{\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right)}_{\text{Chain Rule}} + \frac{\partial^2 \phi}{\partial z^2}$$

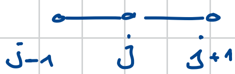


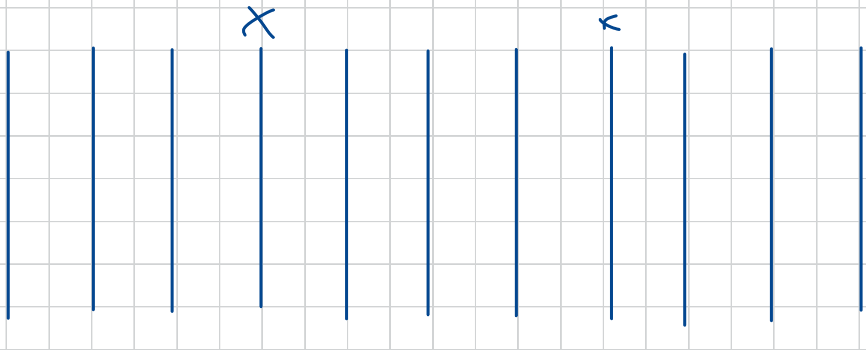
$$\frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r}\frac{\partial \phi}{\partial r} + \frac{\partial^2 \phi}{\partial z^2}$$



Dirichlet: $\phi(0) = \phi_0$

Von Neumann: $\phi'(0) = 0$

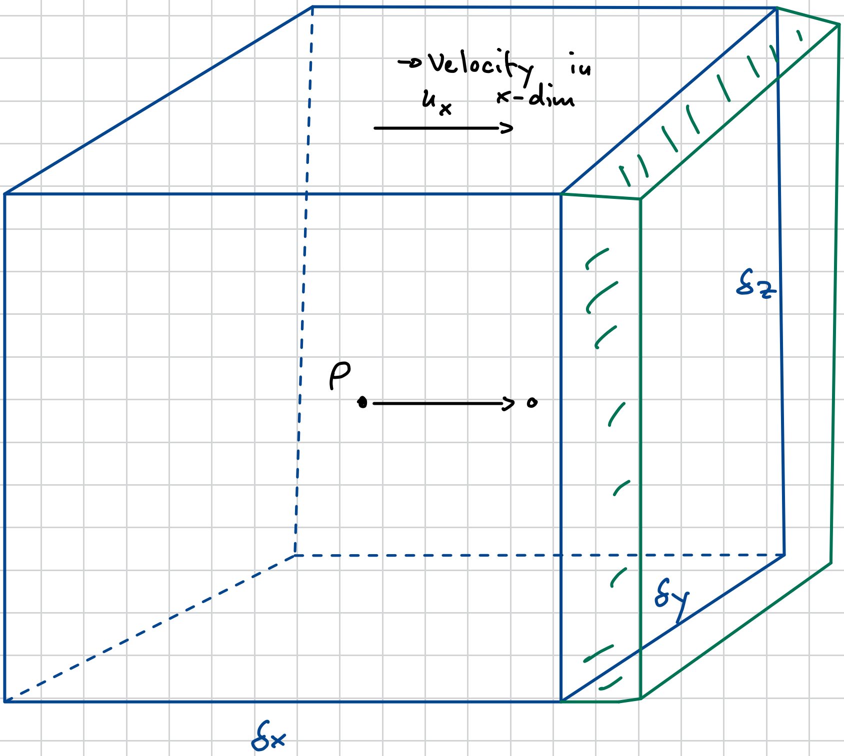




Hypobolic PDE's

↳ Time debendent PDE's

↳ Conseved quantity e.g. Ink in wate → We want to observe how it behaves is space & time



→ small volume δV

→ how much "wate" leaves the cube

Net loss: Material leaving the cube

$$\boxed{\rho_w \cdot \Delta x \, \delta y \, \delta z}$$

Per unit t:

$$\Big(\rho_w \underbrace{\frac{\Delta x}{t}}_{u_x}\Big) \delta y \, \delta z$$

$$\rho_w \cdot u_x = \underline{\underline{\text{Flux}}}$$

Define $v := \langle u_x, u_y, u_z \rangle$

$$\rho_w = \rho(x + \tfrac{1}{2}\delta x, t) \cong \rho(x,t) + \frac{1}{2}\delta x \frac{\partial \rho}{\partial x}\Big|_{x,t}$$

→ Flux through surface is given by:

$$\rho \cdot u(x + \tfrac{1}{2}\delta x, t) \cong \rho u(x,t) + \frac{1}{2}\delta x \frac{\partial(\rho u)}{\partial x}\Big|_{x,t}$$

$A: \left\{ \rho u + \frac{1}{2}\delta x \frac{\partial(\rho u)}{\partial x} \right\} \delta y \, \delta z$ → Material leaving Cube

$B: \left\{ \rho u - \frac{1}{2}\delta x \frac{\partial(\rho u)}{\partial x} \right\} \delta y \, \delta z$ → Material entering Cube

$$\frac{\partial \rho}{\partial t} \cdot \underbrace{\cancel{\delta V}}_{\delta x \delta y \delta z} = B - A = -\cancel{\delta x}\frac{\partial(\rho u)}{\partial x}\cancel{\delta y}\cancel{\delta z}$$

$$\frac{\partial \rho}{\partial t} = -\frac{\partial(\rho u_x)}{\partial x}$$

→ when movement only in x-direction

$$= -\frac{\partial(\rho u_y)}{\partial y}$$

$$= -\frac{\partial(\rho u_z)}{\partial z}$$

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \vec{u})$$

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0}$$

↳ Hypobolic PDE

↳ Principle of conservation

Vector

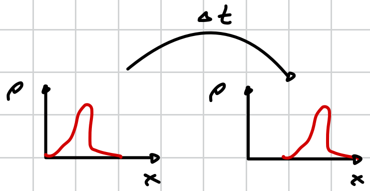
$$\nabla \cdot (\vec{A}) := \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

→ If we have: (1) Conservation of Mass
(2) Conservation of Energy
(3) Conservation of Momentum

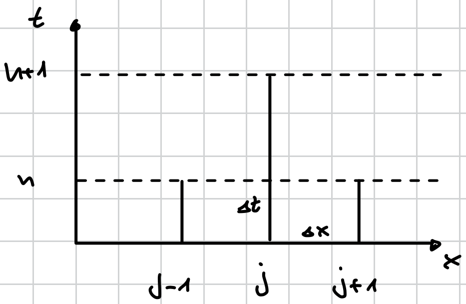
$\Big\}$ Euler Equations

we get the Equations of Hydrodynamics without viscosity.

→ Linear Advection Equation :



$$\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} = 0$$



$$\left.\frac{\partial \rho}{\partial t}\right|_j \cong \frac{\rho_j^{n+1} - \rho_j^n}{\Delta t}$$

$$\left.\frac{\partial \rho}{\partial t}\right|_j^n \cong \frac{\rho_{j+1}^n - \rho_{j-1}^n}{2\Delta x}$$

$$\boxed{\rho_j^{(n+1)} = \rho_j^{(n)} - \tfrac{1}{2} c\left(\rho_{j+1}^{(n)} - \rho_{j-1}^{(n)}\right) \qquad c = \frac{\Delta t \cdot u}{\Delta x}}$$

→ Method is absolutely unstable

→ Von Neuman stability Analysis (Our Function should not grow)

$$\rho_j^{(n)} = A^n e^{ij\theta}$$

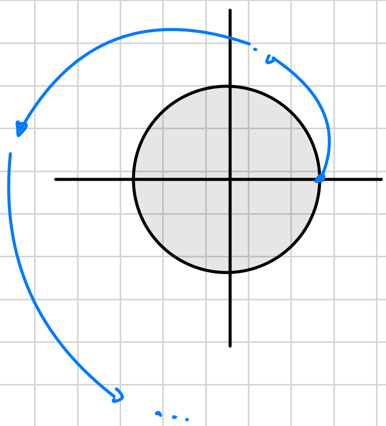
$$A^{n+1} e^{ij\theta} = A^n e^{ij\theta} - \tfrac{1}{2} c\left(A^n e^{ij\theta} - A^n e^{i(j-1)\theta}\right)$$

→ Cancel $A^n e^{ij\theta}$

$$A = 1 - c\left(\boxed{\frac{e^{i\theta} - e^{-i\theta}}{2}}\right) \qquad \boxed{\frac{e^{i\theta} - e^{-i\theta}}{2}} = i \sin\theta$$

$$A = 1 - c \sin\theta$$

$$|A|^2 \equiv A * A = 1 + c^2 \sin^2\theta \quad \begin{cases} |A|^2 > 1 \\ \forall\, c, \theta \in \mathbb{R} \end{cases}$$



→ No matter the wave → the function is not stable

→ Computational Method is useless

Simple Modification (LAX Method)

Replace $\rho_j^{(n)} \to \frac{1}{2}\left(\rho_{j+1}^{(n)} + \rho_{j-1}^{(n)}\right)$ → Avg. of neighbours

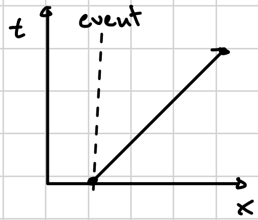
$$\rho_j^{(n+1)} = \frac{1}{2}\left(\rho_{j+1}^{(n)} + \rho_{j-1}^{(n)}\right) - \frac{1}{2} c \left(\rho_{j+1}^{(n)} - \rho_{j-1}^{(n)}\right)$$

Von Neuman :

$$A = \cos\theta - ic \sin\theta$$

$$|A|^2 = \cos^2\theta + \sin^2\theta \overset{?}{>} 1$$

→ Stable if $|c| < 1$ → Courant Condition $\boxed{\frac{\Delta t}{\Delta x}} \cdot u < 1$ → velocity



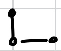
Grid velocity

$\frac{u}{u_{grid}} < 1$

physical velocity

Grid velocity
→ fastest information can travel on grid

Upwind Method:

Stencil :  or 

$$\rho_j^{(n+1)} = \rho_j^{(n)} - c\left(\rho_j^{(n)} - \rho_{j-1}^{(n)}\right) \quad \text{if } u > 0$$

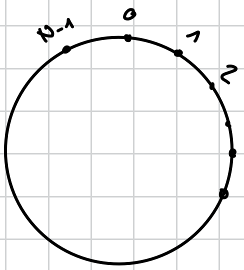
$$= \rho_j^{(n)} - c\left(\rho_{j+1}^{(n)} - \rho_j^{(n)}\right) \quad \text{if } u < 0$$

1st order Upwind

LAX - Wendroff Method

$$\rho_{j+1}^{(n)} = \frac{1}{2} c(1+c)\rho_{j-1}^{(n)} + (1-c^2)\rho_j^{(n)} - \frac{1}{2} c(1-c)\rho_{j+1}^{(n)} \quad \text{for } u > 0$$

→ Particle leaving the grid reenters from other side
↳ behaviour can be visualized as a circle



ρ

→ In perfect computation the "Ducks" should go around forever → Not happening with our solution

0. → "Loose" Method
1. → LAX
2. → 1st order Upwind
3. → LAX Wendroff

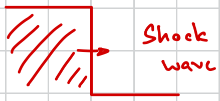
} for $c > 1$, $c < 1$, $c = 1$

Finite Volume

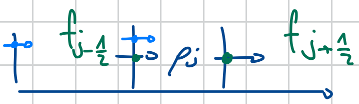
$$\frac{\partial \rho}{\partial t} + a \frac{\partial \rho}{\partial x} = 0$$

approximate the derivatives

→ Integral Equations are more natural to describe conservation Laws



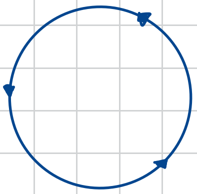
↳ derivative is infinite or even undefined



$$\rho_j^{n+1} = \rho_j^n + \frac{\Delta t}{\Delta x}\left[\underbrace{f_{j-\frac{1}{2}}} - \underbrace{f_{j+\frac{1}{2}}}\right]$$

approximate these fluxes

If all fluxes are added up in a circle



$$\sum_{i=0}^{N}(f_i) = 0$$

Finite Volume Methods

→ Characteristics

↳ a = constant = slope

↳ "Way how information flows"

$$\frac{\partial \rho}{\partial t} + a \frac{\partial \rho}{\partial x} = 0$$

→ Courant Number = c

$$\rho_j^{n+1} = \overbrace{\boxed{\frac{a \cdot \Delta t}{\Delta x}}}^{\text{"Ca"}} \rho_{j-1}^n + \boxed{\left(1 - \frac{a \cdot \Delta t}{\Delta x}\right)} \rho_j^n$$

→ $\rho_j^{n+1} = c\rho_{j-1}^n + \rho_j^n - c\rho_j^n$

$$\rho_j^{n+1} - \rho_j^n + c\left(\rho_j^n - \rho_{j-1}^n\right) = 0$$

→ first order Upwind (CIR-Method)

Loose Method:

$$\frac{\rho_j^{n+1} - \rho_j^n}{\Delta t} + a\frac{\rho_{j+1}^n - \rho_{j-1}^n}{2\Delta x} = 0$$

→ Taylorexpand ρ in time to 2nd order

$$\rho_j^{n+1} = \rho_j^n + \Delta t\left(\frac{\partial \rho}{\partial t}\right) + \frac{\Delta t^2}{2}\left(\frac{\partial^2 \rho}{\partial^2 t}\right)$$

→ Taylorexpand ρ in space to 2nd order

$$\rho_{j+1}^n = \rho_j^n + \Delta x\left(\frac{\partial \rho}{x}\right) + \frac{\Delta x^2}{2}\left(\frac{\partial^2 \rho}{\partial^2 x}\right) \quad / \quad \rho_{j-1}^n = \rho_j^n + \Delta x\left(\frac{\partial \rho}{x}\right) + \frac{\Delta x^2}{2}\left(\frac{\partial^2 \rho}{\partial^2 x}\right)$$

$$\frac{\cancel{\Delta t}\left(\frac{\partial \rho}{\partial t}\right) + \frac{\Delta t^{\cancel{2}}}{2}\left(\frac{\partial^2 \rho}{\partial t^2}\right)}{\cancel{\Delta t}} + a \cdot \left(\frac{\cancel{2\Delta x}\left(\frac{\partial \rho}{\partial x}\right)}{\cancel{2\Delta x}} \right) = 0$$

$$\frac{\partial \rho}{\partial t} + a \frac{\partial \rho}{\partial x} = \underbrace{- \frac{\Delta t}{2}\left(\frac{\partial^2 \rho}{\partial t^2}\right)}_{\text{Error} \to \text{diffusion Term}} \quad \dots \text{ higher order terms}$$

↳ negative diffusion ⇒ unstable ⚡

$$\frac{\partial \rho}{\partial t} + a \frac{\partial \rho}{\partial x} = - a^2 \frac{\Delta t}{2} \frac{\partial^2 \rho}{\partial x^2}$$ Modified Equation → Advection Diffusion

---

Exercise: try disect other method (Upwind ...)

2-D Advection:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) = 0$$ where $\vec{u} = \langle a, b \rangle$ $\quad a, b > 0$

First order finite difference is:

$$\frac{\rho_{j\ell}^{n+1} - \rho_{j\ell}^{n}}{\Delta t} + a \frac{\rho_{j\ell}^{n} - \rho_{j-1,\ell}^{n}}{\Delta x} + b \frac{\rho_{j\ell}^{n} - \rho_{j\ell-1}^{n}}{\Delta y} = 0$$

$$\rho_{j,\ell}^{n+1} = \rho_{j\ell}^{n} - \Delta t \left(a \left(\frac{\rho_{j\ell}^{n} - \rho_{j-1,\ell}^{n}}{\Delta x} \right) + b \left(\frac{\rho_{j\ell}^{n} - \rho_{j\ell-1}^{n}}{\Delta y} \right) \right)$$

→ Look at stability: (Von Neumann e.g.)

$C_a > 0$

$C_b > 0$

$$C_a = \frac{a \Delta t}{\Delta x}$$

$$\frac{a \Delta t}{\Delta x} + \frac{b \Delta t}{\Delta y} \leq 1$$

→ $C_a + C_b \leq 1$ → Sum of Courant Numbers

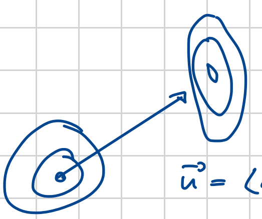
→ Modified Equation:

$$\frac{\partial \rho}{\partial t} + a \frac{\partial \rho}{\partial x} + b \frac{\partial \rho}{\partial y} = \frac{a \Delta x}{2}(1 - C_a) \frac{\partial^2 \rho}{\partial x^2} + \frac{b \Delta y}{2}(1 - C_b)\frac{\partial^2 \rho}{\partial y^2} - \boxed{ab\, \Delta t \frac{\partial^2 \rho}{\partial x \partial y}}$$

Cross Derivative

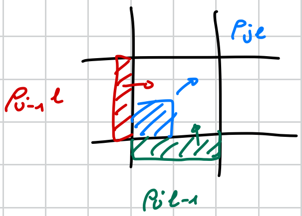
→ Don't preserve Shape ⚡

↳ if we difuse in tangentially → we antidifuse in direction of Movement

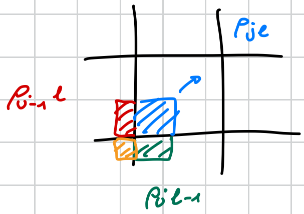


$\vec{u} = \langle a, b \rangle$

Corner Transport Upwind scheme :



→ No info from last cell ?



$$\rho^{n+1}_{jl} = \boxed{(1-C_a)(1-C_b)\rho^n_{j,l}} + \boxed{C_a(1-C_b)\rho^n_{j-1,l}} + \boxed{(1-C_a)\,C_b\,\rho^n_{j,l-1}} + \boxed{C_a\,C_b\,\rho^n_{j-1,l-1}}$$

→ Stability : $0 \le C_a \le 1,\quad 0 \le C_b \le 1$

→ Able to write in two steps (1) horizontal → (2) vertical

Temporary State

$$\rho^*_{jl} = (1-C_a)\rho^n_{jl} + C_a\,\rho^n_{j-1,l}$$

$$\rho^{n+1}_{j,l} = (1-C_b)\rho^*_{jl} + C_b\,\rho^n_{j,l-1}$$

1-D Hydrodynamics

↳ Conservation of Mass / Energy / Momentum
(scalar, scalar, vector)

LAX-Method:

$$\rho_i^{n+1} = \frac{1}{2}\left(\rho_{i+1}^n + \rho_{i-1}^n\right) - \frac{c}{2}\left(\rho_{i+1}^n - \rho_{i-1}^n\right) \quad , \quad c = \frac{u\Delta t}{\Delta x}$$

→ For linear advection: $F(\rho) = \rho \boxed{u}$ constant

↳ rewrite Equation using fluxes:

$$\rho_i^{n+1} = \frac{1}{2}\left(\rho_{i+1}^n + \rho_{i-1}^n\right) - \frac{\Delta t}{2\Delta x}\left(F_{i+1}^n - F_{i-1}^n\right)$$ } Flux-version of LAX

A

First Order Upwind: (Flux version)

$$\rho_i^{n+1} = \rho_i^n - \frac{\Delta t}{\Delta x}\left(F_i^n - F_{i-1}^n\right), \quad u \geq 0$$

$$\rho_i^{n+1} = \rho_i^n - \frac{\Delta t}{\Delta x}\left(F_{i+1}^n - F_i^n\right), \quad u \leq 0$$

A')

→ To do Hydrodinamics we just need to use the appropriate Flux

1. Conservation of Mass: ρ, → $F(\rho) = \rho \boxed{u}$ variable
2. Conservation of Energy: E, → $F(u) = u(E+P)$
3. Conservation of Momentum: ρu → $F(u) = \rho u^2 + P$

→ State Vector:

$$u = \begin{pmatrix} \rho \\ \rho u \\ E \end{pmatrix} \qquad F = \begin{pmatrix} \rho u \\ \rho u^2 + P \\ u(E+P) \end{pmatrix}$$ } Don't behave like vectors

$$\frac{\partial u}{\partial t} + \nabla \cdot F(u) = 0$$ → conservation of Stuff (u)

1-D $\frac{\partial}{\partial x}$

↳ 3 Partial differential Equations

↳ 4 Unknowns $\rho, \rho u, E, P$!

↳ $E = \underbrace{\frac{1}{2}\rho u^2}_{\text{Movement}} + \underbrace{\rho e}_{\text{Thermal}}$ → "specific" thermal energy density per unit Mass

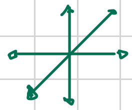
↳ 4 Unknowns $\rho, \rho u, e, P$

→ Equation of State: $PV = nRT$ (Ideal Gas Equation)

Pressure, density, Temperature

$\rightarrow \gamma = \frac{f+2}{f}$ $\qquad e = \frac{p}{(\gamma-1)\rho}$ $\qquad$ (Also Ideal Gas)

f = deg of Freedom

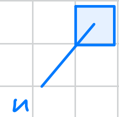


In 3-D $\quad f = 3$

1-D : $\qquad \gamma = 3$

$$e = \frac{1}{2}\frac{p}{\rho} \qquad E = \frac{1}{2}\rho u^2 + \frac{1}{2} p$$

$$F(u) = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u\left(\frac{1}{2}\rho u^2 + \frac{3}{2} p\right) \end{pmatrix}$$



Finite Volumes

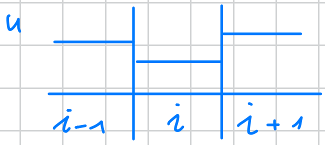
→ Averag values of the "material" in the Cell

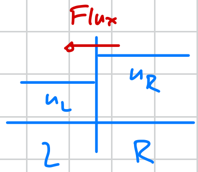
$$\langle u \rangle_i^n = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} u(x, t^n)\, dx$$

→ This is at one instace in Time, one can also conside the average over a timestep

other option: (Midpoint)

$$\frac{\langle u \rangle_i^{n+1} - \langle u \rangle_i^n}{\Delta t} + \frac{F_{i+\frac{1}{2}}^{n+\frac{1}{2}} - F_{i-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} = 0$$

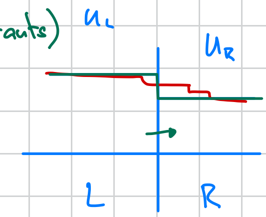




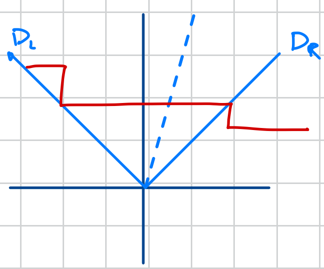
→ Gudunof Methods (Piecewise Constants)

→ Rieman Solve

$$F(U_L, U_R)$$



→ Approximate the Flux



C_s = Soundspeed

$= \sqrt{\gamma \frac{P}{\rho}}$ speed of sound

$D_L = u - C_s$

$D_R = u + C_s$

$D_{Max} = \max(D_L, D_R) = |u| + C_s$

→ Lax Friedrichs Riemann Solver

$$\boxed{F^n_{i-\frac{1}{2}} = \frac{1}{2}\left[\underset{\uparrow\, u^n_i}{F^n_i} + \underset{\uparrow\, u^n_{i-1}}{F^n_{i-1}}\right] - \frac{1}{2} D_{Max}\left[u^n_i - u^n_{i-1}\right]}$$

→ Method A : (Really bad)

At cell center

$$u^{n+1}_i = \frac{1}{2}\left(u^n_{i+1} + u^n_{i-1}\right) - \frac{\Delta t}{2\Delta x}\left(F^n_{i+1} - F^n_{i-1}\right)$$ → Not conservative!

$$F = \begin{pmatrix} \rho u \\ \rho u^2 + P \\ u\left(\frac{1}{2}\rho u^2 + \frac{3}{2}P\right) \end{pmatrix}$$

→ Method B

At Faces

$$\langle u \rangle^{n+1}_i = \langle u \rangle^n_i - \frac{\Delta t}{\Delta x}\left[F^n_{i+\frac{1}{2}} - F^n_{i-\frac{1}{2}}\right]$$ → Conservative

→ Method C (Predictor Corrector)

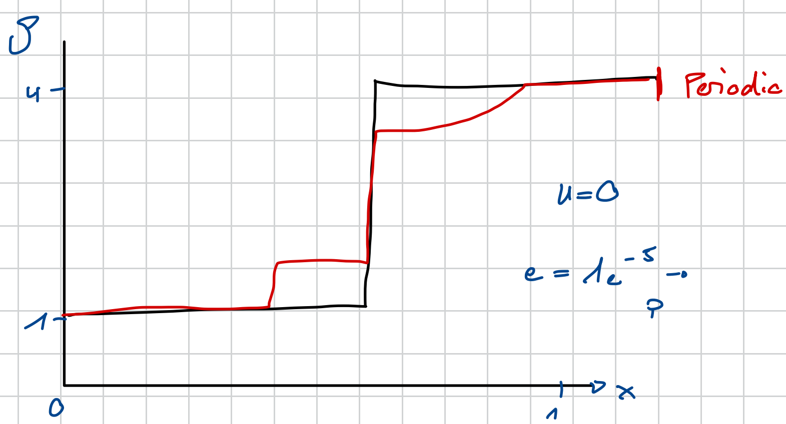
Predictive step

1. $$u_i^{\text{"}n+\frac{1}{2}\text{"}} = u^n_i - \frac{\Delta t}{2\Delta x}\left[F^n_{i+\frac{1}{2}} - F^n_{i-\frac{1}{2}}\right]$$ → Method B for a half time step

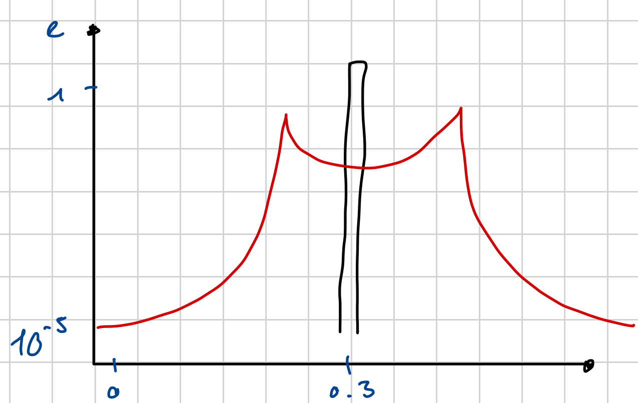
2. $$u^{n+1}_i = u^n_i - \frac{\Delta t}{\Delta x}\left[\underbrace{F^{n+\frac{1}{2}}_{i+\frac{1}{2}}}_{F(u^*_{i+1},\, u^*_i)} - \underbrace{F^{n+\frac{1}{2}}_{i-\frac{1}{2}}}_{F(u^*_i,\, u^*_{i-1})}\right]$$

Assignment :

Sod- Shock Tube



Shedov Taylor Blastwave



ESC401 → ESC402
Paralell Computing

Parallel Computing

→ Nature is paralell → why shoud computing be seriel

→ # of Transistors $\simeq 1.6^{year}$ → 10 years $\simeq$ factor of 100